1. Coulomb's Law & Unit of Charge

Definition: Coulomb's Law states that the electrostatic force of attraction or repulsion between two stationary point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.

Electrostatics deals with the study of electric charges at rest. The fundamental interaction between these stationary charges is described by Coulomb's law. Like charges (positive-positive or negative-negative) repel each other, whereas unlike charges (positive-negative) attract each other.

Mathematical Expression:

Let two point charges $\mathbf{q_1}$ and $\mathbf{q_2}$ be separated by a distance **r** in a vacuum. The force **F** between them is:

- $F \propto q_1 \times q_2$ (Directly proportional to the product of charges)
- $F \propto 1 / r^2$ (Inversely proportional to the square of distance)

$$\mathbf{F = k \cdot (|q_1| \cdot |q_2|) / r^2}$$

Here, **k** is the electrostatic constant. In SI units, for vacuum/air, $k = 1 / (4\pi\varepsilon_0) = 9 \times 10^9$ $N \cdot m^2/C^2$. (where ε_0 is the permittivity of free space).

Unit of Charge (Coulomb):

The SI unit of electric charge is the **Coulomb (C)**.
Definition of 1 Coulomb: One Coulomb is that amount of charge which, when placed at a distance of 1 meter from an identical charge in a vacuum, experiences a repulsive force of exactly 9×10^9 Newtons.

```
+ q₁ ------------------------ r ------------------------ + q₂
---> F (Repulsion)                              F (Repulsion) <---

+ q₁ ------------------------ r ------------------------ - q₂
<--- F (Attraction)                             F (Attraction) --->
```

👉 **Hinglish Explanation:** Do static (ruke hue) charges ke beech lagne wala force (chahe attraction ho ya repulsion) un charges ki value ke multiply ke directly proportional aur unke beech ki doori ke square ke inversely proportional hota hai. Iski SI unit Coulomb (C) hoti hai. Ek dam Newton's law of gravitation jaisa formula hai!

2. Electric Field & Electric Lines of Force

Definition: An electric field is the space or region around an electric charge within which its electrical force of attraction or repulsion can be felt by another test charge.

Electric Field Intensity (E):

The electric field intensity at any point is defined as the force experienced by a unit positive test charge placed at that point. It is a vector quantity.

$E = F / q_0$ (Unit: Newton/Coulomb or N/C)

Electric Lines of Force:

An electric line of force is an imaginary smooth curve drawn in an electric field along which a free, isolated unit positive charge would move if allowed to do so. The tangent to this curve at any point gives the direction of the electric field at that point.

Properties of Electric Lines of Force:

- They always originate from positive charges and terminate at negative charges.
- **No two lines of force can ever intersect each other.** If they did, there would be two tangents at the point of intersection, meaning two directions of the electric field at a single point, which is impossible.
- They are continuous, smooth curves without any breaks in a charge-free region.
- They do not form closed loops (unlike magnetic lines of force).
- They always enter or leave a charged conductor normally (at right angles/90 degrees) to its surface.
- The closer the lines, the stronger the electric field in that region.

```
Positive Charge (+): Lines go OUT       Negative Charge (-): Lines go IN
       ^   |   ^                                \   |   /
        \  |  /                                  \  |  /
    <---- (+) ---->                          ----> (-) <----
        /  |  \                                  /  |  \
       v   |   v                                /   |   \
```

👉 **Hinglish Explanation:** Electric Field matlab kisi charge ka 'ilaqa' jaha tak uska asar mehsus hota hai. Lines of force imaginary lines hain jo batati hain ki ek +ve test charge kidhar jayega. Ye hamesha + se nikal kar - par khatam hoti hain. Do lines kabhi ek dusre ko nahi kaatti (intersect nahi karti), ye board exam ka sabse famous question hai!

3. Electric Flux, Potential & Potential Difference

Definition: Electric Flux (Φ) is the total number of electric lines of force passing normally

(perpendicularly) through a given surface area. Electric Potential is the work done in bringing a unit positive charge from infinity to a specific point against the electrostatic force.

Electric Flux (Φ):

It measures the flow of the electric field through an area. Mathematically, it is the dot product of the Electric Field vector (E) and the Area vector (A).

$$\Phi = E \cdot A = E\,A\cos(\theta)$$
(Unit: $N \cdot m^2/C$ or Volt·meter)

Where θ is the angle between the electric field lines and the normal (perpendicular) to the surface.

Electric Potential (V):

It represents the electrical potential energy per unit charge. If *W* amount of work is done in bringing a test charge q_0 from infinity to a point, then potential V at that point is:

$$V = W / q_0 \text{ (Unit: Joules/Coulomb or Volt)}$$

It is a scalar quantity. The potential of the Earth is considered to be zero.

Potential Difference (ΔV):

It is the work done in moving a unit positive charge from one point to another within an electric field.
$\Delta V = V_2 - V_1 = W(1 \text{ to } 2) / q_0$. Current flows only when there is a potential difference between two points.

> 👉 **Hinglish Explanation:** Flux (Φ) ka matlab hai kisi area se kitni 'electric lines' cross kar rahi hain (jaise khidki se hawa aana). Electric Potential (Volt) ka matlab hai kisi charge ko infinity (bahut door) se kisi point tak laane me kitni mehnat (work) lagi. Potential difference wahi voltage hai jo battery me hoti hai, jiske karan current flow karta hai.

4. Gauss's Law & Its Applications

Definition: Gauss's Law states that the total electric flux (Φ) passing through any closed hypothetical surface (Gaussian surface) in free space is equal to $1/\varepsilon_0$ times the net charge (q) enclosed by that surface.

$$\oint E \cdot ds = q / \varepsilon_0$$

Gauss's law is a powerful tool to calculate electric fields for highly symmetric charge distributions, replacing complex calculus integrations.

Applications of Gauss's Law:

1. **Electric Field due to a Straight Infinitely Long Charged Conductor:**
   Consider a long straight wire with linear charge density λ (charge per unit length). We assume a cylindrical Gaussian surface of radius 'r' and length 'l' around it.

Formula: **$E = \lambda / (2\pi\varepsilon_0 r)$**
Observation: E is inversely proportional to r ($E \propto 1/r$).

2. **Electric Field due to an Infinite Plane Charged Sheet:**
Consider a thin plane sheet with surface charge density σ (charge per unit area). We assume a cylindrical pillbox as the Gaussian surface passing through the sheet.
Formula: **$E = \sigma / (2\varepsilon_0)$**
Observation: E is uniform and independent of the distance 'r' from the sheet.
3. **Electric Field due to a Uniformly Charged Spherical Shell (Sphere):**
Let a spherical shell of radius 'R' have a total charge 'q'.
   - **Outside the sphere (r > R):** $E = 1/(4\pi\varepsilon_0) \cdot (q/r^2)$. (Behaves as if all charge is concentrated at the center).
   - **On the surface (r = R):** $E = 1/(4\pi\varepsilon_0) \cdot (q/R^2)$. (Maximum electric field).
   - **Inside the sphere (r < R): E = 0**. (Since charge resides only on the outer surface of a conductor, enclosed charge is zero).

👉 **Hinglish Explanation:** Gauss's law electrostatics ka master formula hai. Ye kehta hai ki kisi bhi closed 3D body se nikalne wala total flux uske andar band charge ka $1/\varepsilon_0$ guna hota hai. Iske applications (straight wire, plane sheet, aur sphere) ke derivations exams me direct 7-8 marks me aate hain. Sphere ke andar Electric field hamesha ZERO hota hai!

5. Capacitor, Working, Types & Capacitance

Definition: A capacitor is a passive electronic component consisting of two conducting surfaces separated by an insulating material (dielectric) used to store electrical energy in the form of an electrostatic field.

Capacitance & its Units:

The ability of a capacitor to store charge is called its Capacitance (C). It is defined as the ratio of the charge (Q) given to a conductor to the rise in its potential (V).

$C = Q / V$
SI Unit: Farad (F)

Since 1 Farad is a very large unit, practical units are microfarad ($1\ \mu F = 10^{-6}$ F) and picofarad ($1\ pF = 10^{-12}$ F).

Working Principle of a Capacitor:

The principle is based on artificially increasing the capacitance of a conductor. When an uncharged, earthed conductor is brought near a charged conductor, the potential of the charged conductor decreases due to the induced opposite charge on the closer face of the earthed conductor. Since C = Q/V, a decrease in potential (V) causes a massive increase in its capacitance (C) without changing its physical size.

Types of Capacitors:

- **Parallel Plate Capacitor:** Two flat parallel plates separated by a dielectric. (Most common in theory).
- **Spherical Capacitor:** Two concentric spherical shells.
- **Cylindrical Capacitor:** Two coaxial cylinders. (Used in coaxial cables).
- **Electrolytic Capacitor:** Used when high capacitance is required (polarized, has + and - terminals).

- **Ceramic/Paper/Mica Capacitors:** Named after the dielectric material used between the plates.

> 👉 **Hinglish Explanation:** Capacitor (jise aam bhasha me Condenser kehte hain, jaise pankhe me lagta hai) ek aisi device hai jo apne andar electric charge (energy) store karke rakhti hai. Iska basic principle ye hai ki ek charged plate ke paas ek earthing ki hui normal plate laane se pehli plate ka voltage gir jata hai, aur uski charge store karne ki takat (Capacitance) badh jati hai.

6. Parallel Plate Capacitor & Combinations

Definition: A parallel plate capacitor consists of two large plane parallel conducting plates separated by a small distance 'd', with empty space or a dielectric in between.

Capacitance of a Parallel Plate Capacitor:

Let 'A' be the area of each plate and 'd' be the distance between them.

$$C = (\varepsilon_0 \cdot A) / d \text{ (for air/vacuum)}$$

Factors affecting Capacitance:

- Directly proportional to the Area of the plates ($C \propto A$).
- Inversely proportional to the distance between them ($C \propto 1/d$).
- Depends on the nature of the dielectric medium between the plates.

Combination of Capacitors (For Numericals):

Capacitors can be connected in circuits in two primary ways to achieve a desired equivalent capacitance.

1. **Series Combination:**
   - Plates are connected end-to-end.
   - **Charge (Q) is SAME** across all capacitors.
   - **Voltage (V) divides** ($V = V_1 + V_2 + V_3$).
   - Equivalent Capacitance formula is the reciprocal sum (Decreases total capacitance):

$$1/C_{series} = 1/C_1 + 1/C_2 + 1/C_3 + \ldots$$

2. **Parallel Combination:**
   - All positive plates are connected to one point, and all negative to another.
   - **Voltage (V) is SAME** across all capacitors.
   - **Charge (Q) divides** ($Q = Q_1 + Q_2 + Q_3$).
   - Equivalent Capacitance is the direct sum (Increases total capacitance):

$$C_{parallel} = C_1 + C_2 + C_3 + \ldots$$

```
SERIES CONNECTION:            PARALLEL CONNECTION:
   | |      | |      | |             ___  | | ___
---|-|-----|-|-----|-|---         |     | |     |
   | | C₁   | | C₂   | | C₃     ----|___  | | ___|----
                                   | C₁ |  | C₂ |
                                   |___  | | ___|
                                         | | C₃
```

👉 **Hinglish Explanation:** Parallel plate capacitor ki capacity ($C = \varepsilon_0 A/d$) plate ka area badhane se badhti hai aur plates ke beech ki doori kam karne se badhti hai. Dhyan rakhna, capacitors ke Series aur Parallel ka formula Resistors (Class 10th wale) ke ekdum ulta hota hai. Series me capacity kam hoti hai (1/C), aur Parallel me capacity plus hokar badh jati hai (C1+C2).

7. Dielectric, Effect on Capacitance & Breakdown

Definition: A dielectric is an insulating material (like glass, mica, paper, or oil) that does not conduct electricity but can transmit electric effects through electrical polarization.

Effect of Dielectric on Capacitance:

When a dielectric slab is completely inserted between the plates of a parallel plate capacitor, the molecules of the dielectric get polarized. This induced polarization creates an internal electric field that opposes the external electric field, effectively reducing the net potential difference (V) between the plates.
Since C = Q/V, a reduction in V leads to an increase in the capacitance.

$$C_{medium} = K \cdot C_{vacuum}$$

Where **K** is the Dielectric Constant (or relative permittivity ε_r) of the material. Therefore, introducing a dielectric increases the capacitance by a factor of K.

Dielectric Breakdown & Dielectric Strength:

While dielectrics are insulators, they cannot withstand infinitely high voltages.

- **Dielectric Breakdown:** If the electric field applied across a dielectric becomes excessively high, it tears the electrons away from their atoms. The insulating material suddenly becomes a conductor, allowing a huge current to pass, often in the form of a spark. This permanently damages the capacitor.
- **Dielectric Strength:** It is the maximum electric field (or voltage per unit thickness) that a dielectric material can withstand without undergoing breakdown. Unit: kV/mm or V/m.

👉 **Hinglish Explanation:** Dielectric bas ek insulator (jaise kaagaz ya kaanch) hai. Ise capacitor ke plates ke beech daalne se capacitance 'K' times multiply ho kar badh jati hai. Lekin agar aap capacitor ko uski aukaat se zyada voltage doge, to dielectric ke andar ke electrons toot kar flow karne lagenge aur spark hoga, jise 'Dielectric Breakdown' kehte hain (yani capacitor jal gaya).

Quick Revision Notes (Exam Saver)

- **Coulomb's Law:** $F = k(q_1q_2)/r^2$. Like charges repel, unlike attract.
- **Electric Field (E):** Force per unit charge (F/q). Lines never cross each other.
- **Gauss's Law:** Total flux $\Phi = q/\varepsilon_0$. Sphere ke andar Electric field (E) = 0 hota hai.
- **Capacitance Formula:** $C = Q/V$. Parallel plate capacitor: $C = \varepsilon_0 A/d$. Unit: Farad (F).
- **Combinations:** Series me charge (Q) same hota hai ($1/C = 1/C_1 + 1/C_2$). Parallel me voltage (V) same hota hai ($C = C_1 + C_2$).
- **Dielectric:** Insert karne par Capacitance K times badh jati hai. Over-voltage par breakdown ho jata hai.

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